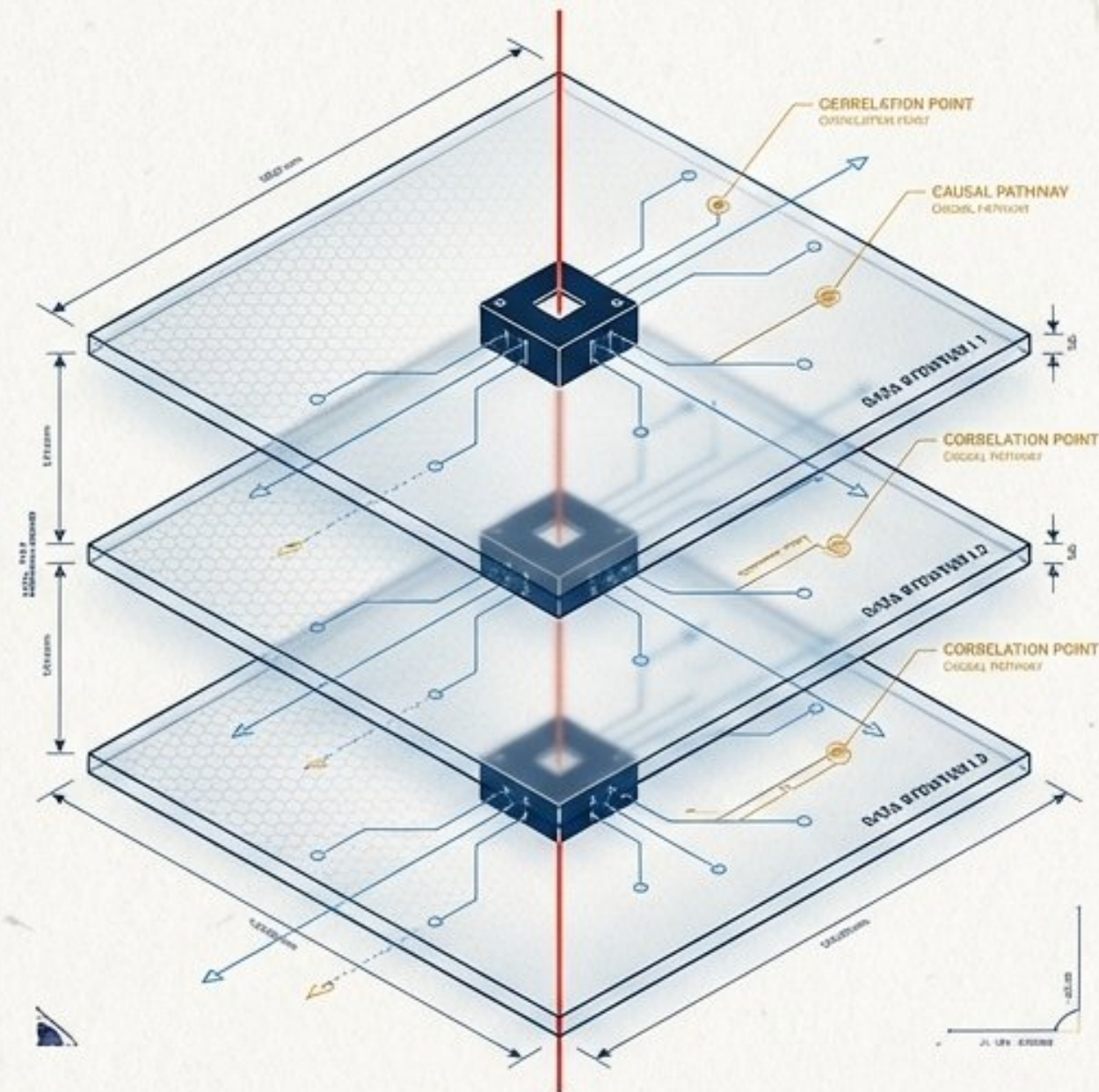




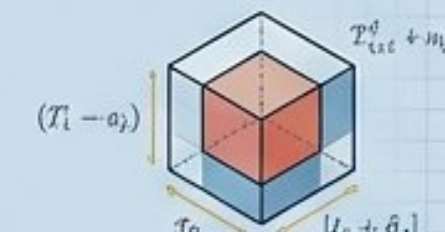
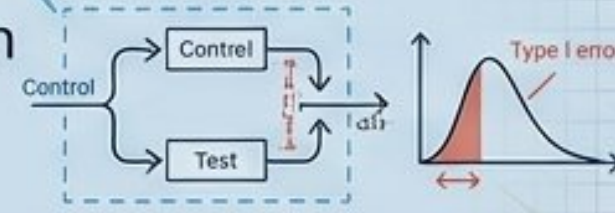
AMACE

Autonomous Multi-Stratum Anomaly Correlation Engine

Domain-agnostic, zero-configuration, counterfactually-validated causal discovery across heterogeneous abstraction layers.

Optimized Patent Specification v2.0
Confidential

Re-engineered for absolute patent defensibility

Prior Art (v1)		AMACE v2
Terminology	Layers / Signals	Strata / Cross-Stratum Causal Tensor 
Data Structure	Ad-hoc JSON	Deviation Signature Tensor (DST) with 7 dimensions 
Dimensional Search	Conceptual MDL	Entropic Relevance Decomposition with $O(k \log D)$ bounds 
Validation	Narrative counterfactuals	Computational Ablation Protocol (CAP) bounding Type I error 

Modern computing operates across three isolated abstraction strata.

S0: Observation Stratum

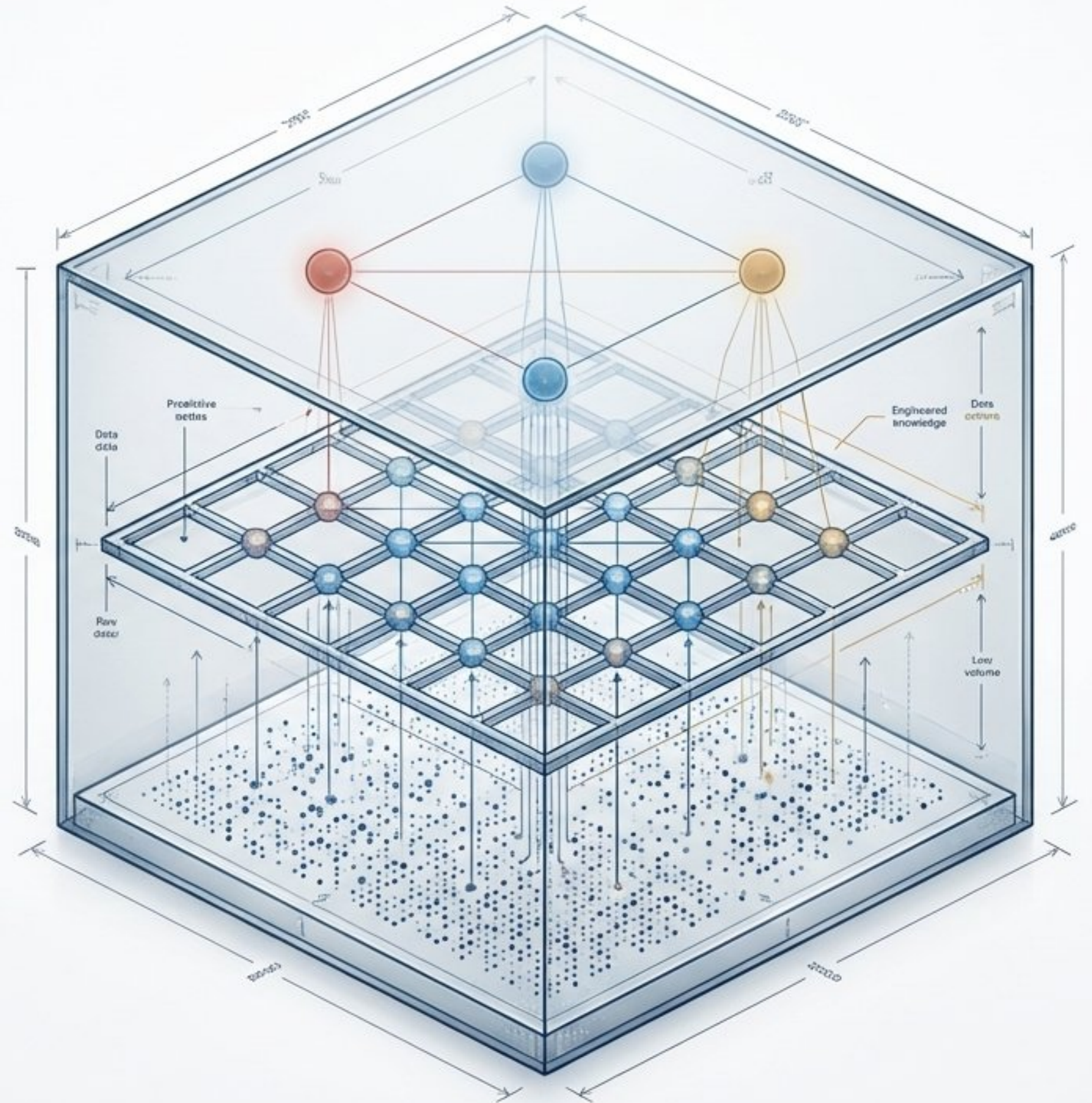
Raw, unprocessed event data.
High volume ($10^6 - 10^{12}$ events/day),
completely unstructured.

S1: Representation Stratum

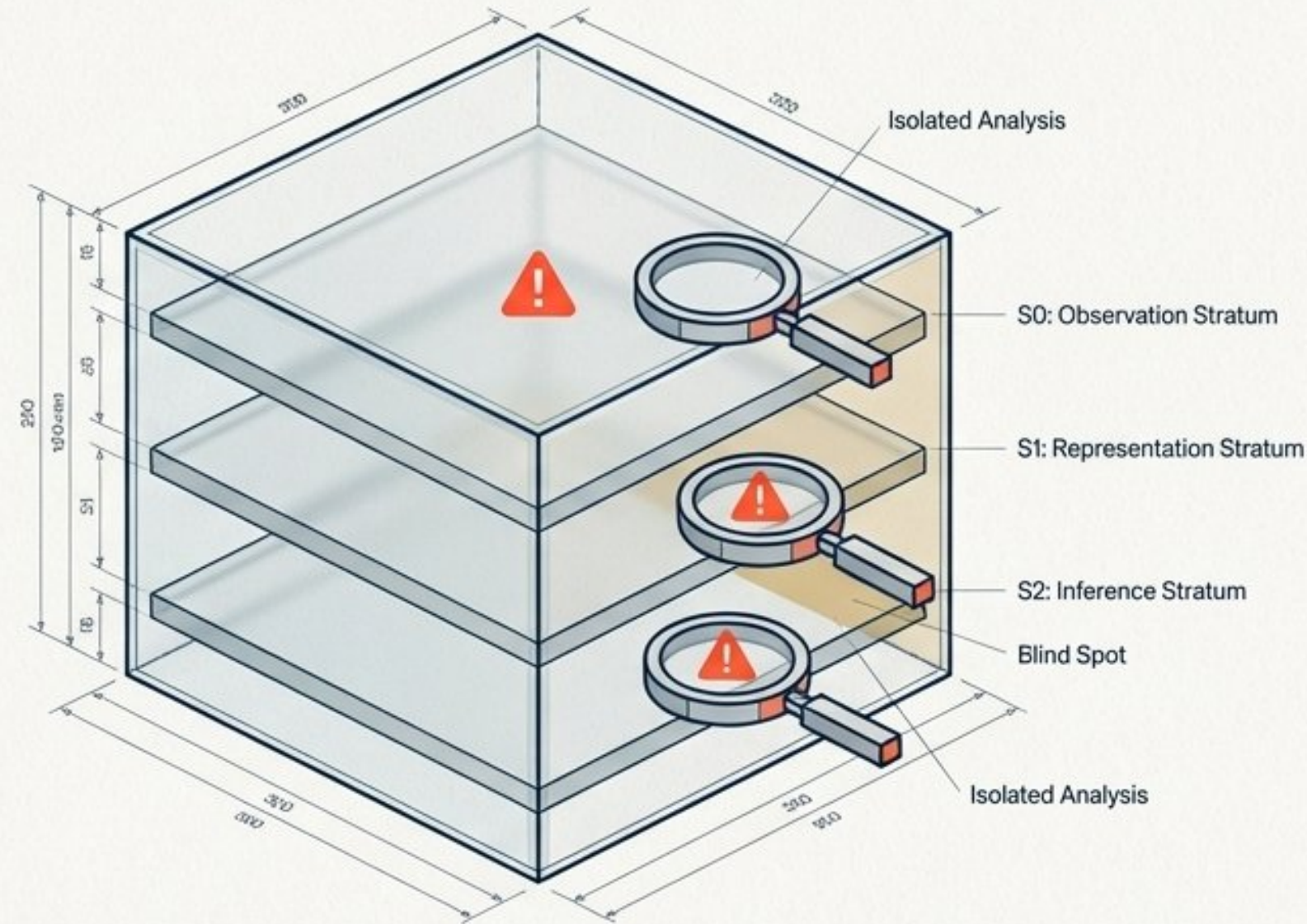
Computed statistical representations.
Moderate volume, highly structured
engineered knowledge.

S2: Inference Stratum

Outputs from statistical or ML models.
Low volume, predictive intelligence.

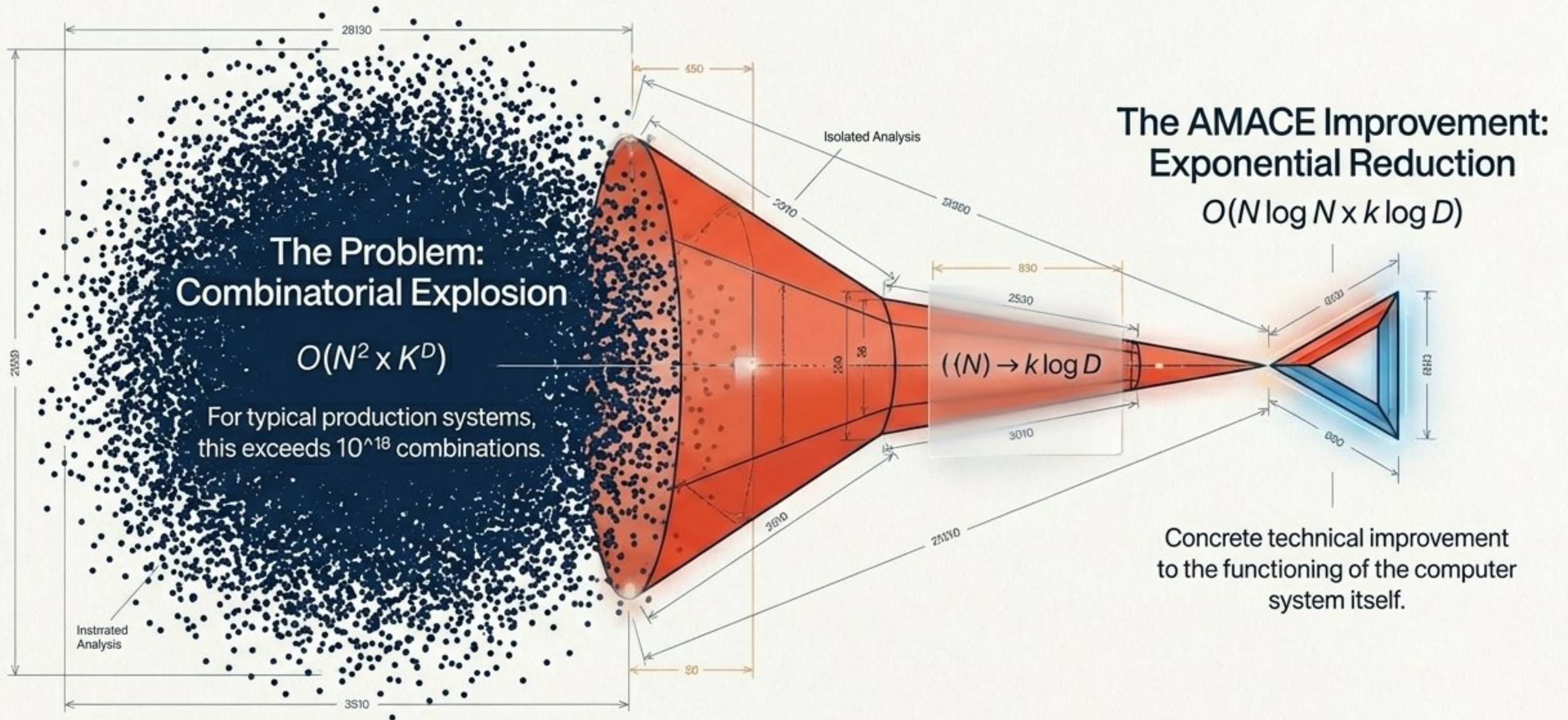


The cross-stratum blind spot conceals shared causal origins.

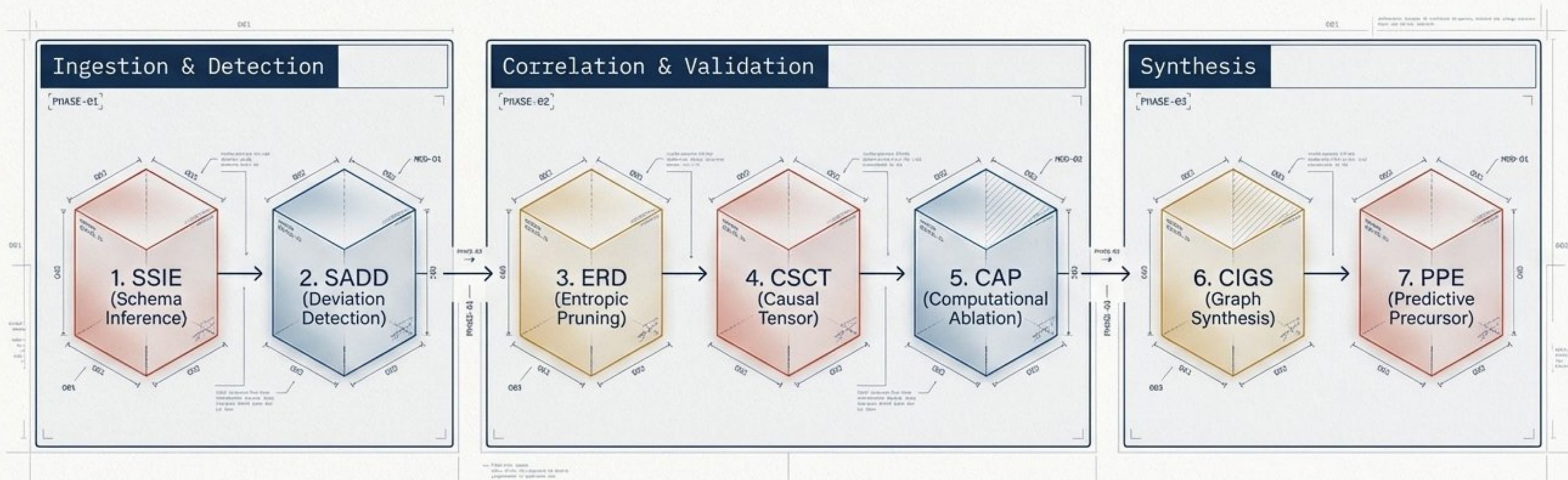


Anomaly detection tools operate strictly within a single stratum. When a system malfunction causes simultaneous deviations across S0, S1, and S2, no existing tool can see the shared origin. They detect the symptoms, but the causal tissue connecting them remains invisible.

Exhaustive cross-stratum correlation is computationally intractable.



A strict, seven-module pipeline for autonomous causal discovery



Autonomous schema inference and universal anomaly representation.

Module 1: SSIE

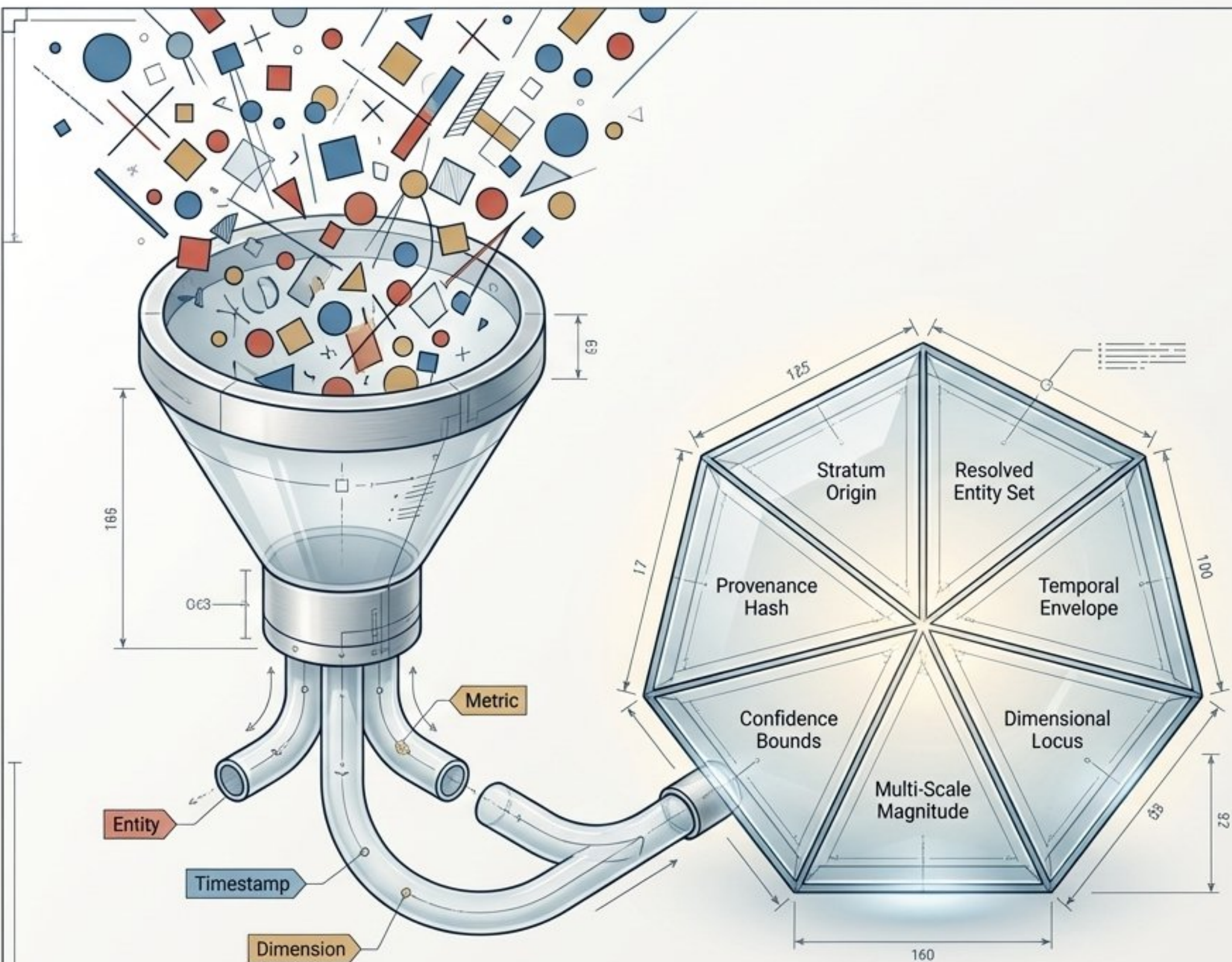
Computes a 4-element classification probability vector (Entity, Timestamp, Dimension, Metric) using Shannon entropy. Zero configuration required.

Module 2: SADD

Converts stratum-specific deviations into a universal Deviation Signature Tensor (DST).

The DST Heptagon:

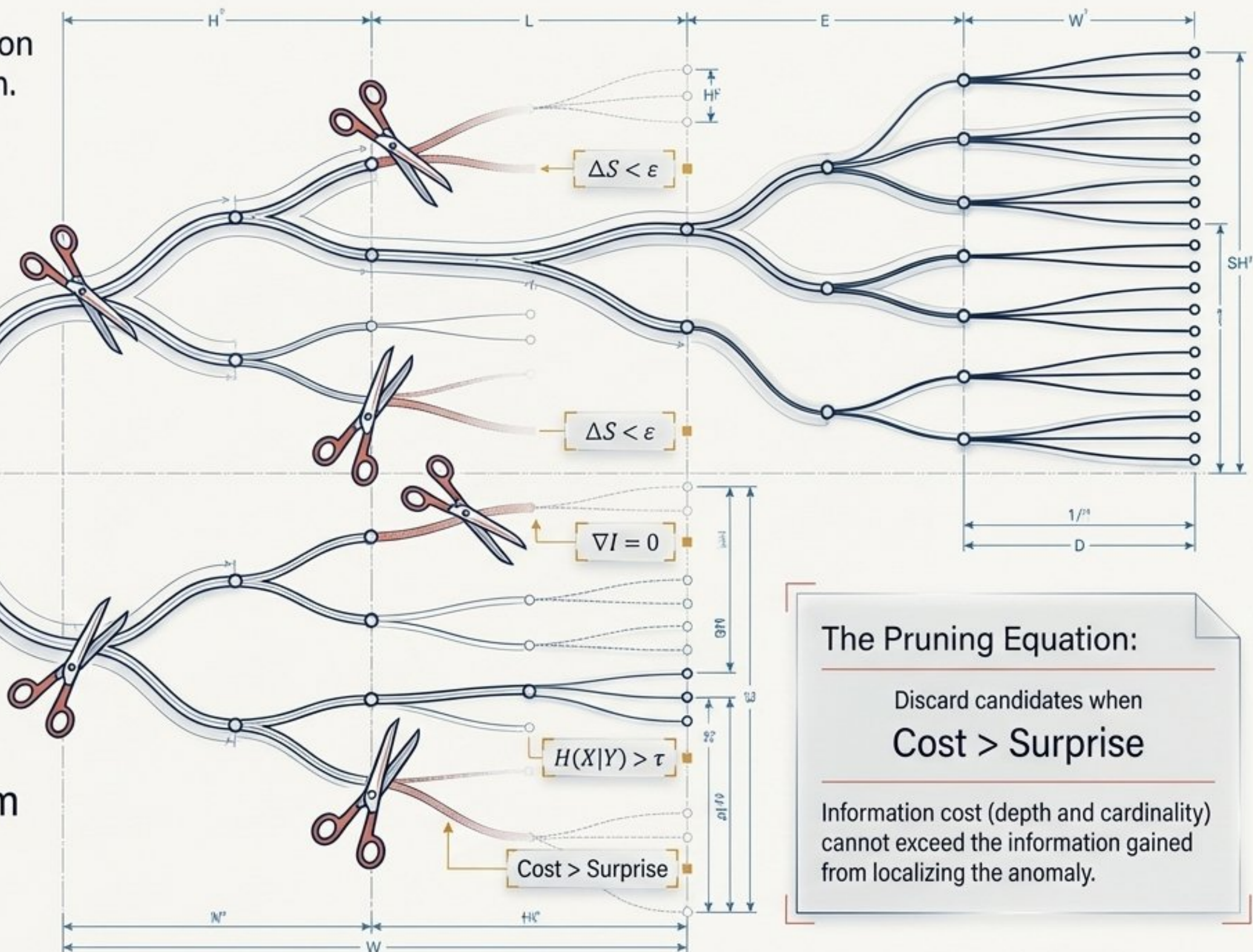
- Stratum Origin
- Resolved Entity Set
- Temporal Envelope
- Dimensional Locus
- Multi-Scale Magnitude
- Confidence Bounds
- Provenance Hash



Entropic Relevance Decomposition isolates the exact sub-population.

Initial Population

AMACE finds the precise dimensional slice of an anomaly using an information-theoretic beam search, achieving 99.97% pruning efficiency.



The Cross-Stratum Causal Tensor measures orthogonal similarities.

α : Entity Coincidence

Set overlap with statistical surprise adjustment.

β : Temporal Precedence

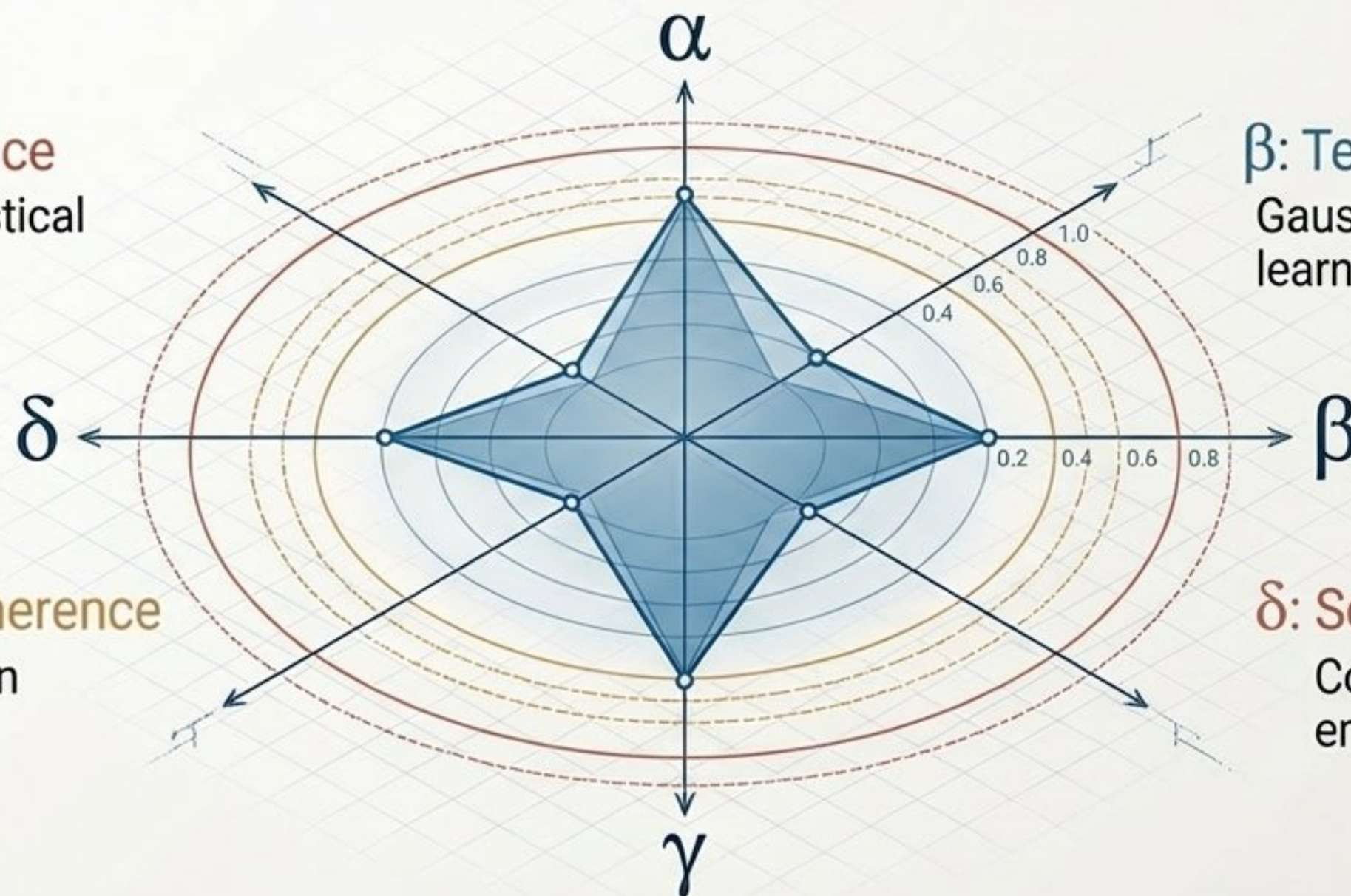
Gaussian kernel mapping learned propagation delays.

γ : Dimensional Coherence

Hierarchical dimension matching.

δ : Semantic Affinity

Contrastive learning embeddings.

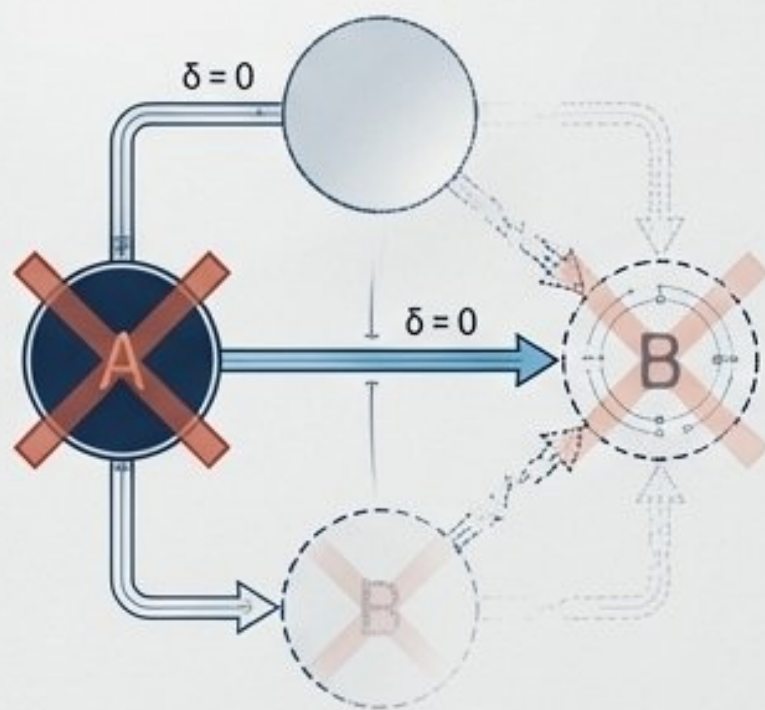


Spectral Normalization: Prevents single-axis dominance; dynamically weights axes based on deviation characteristics.

The Computational Ablation Protocol mathematically proves causality

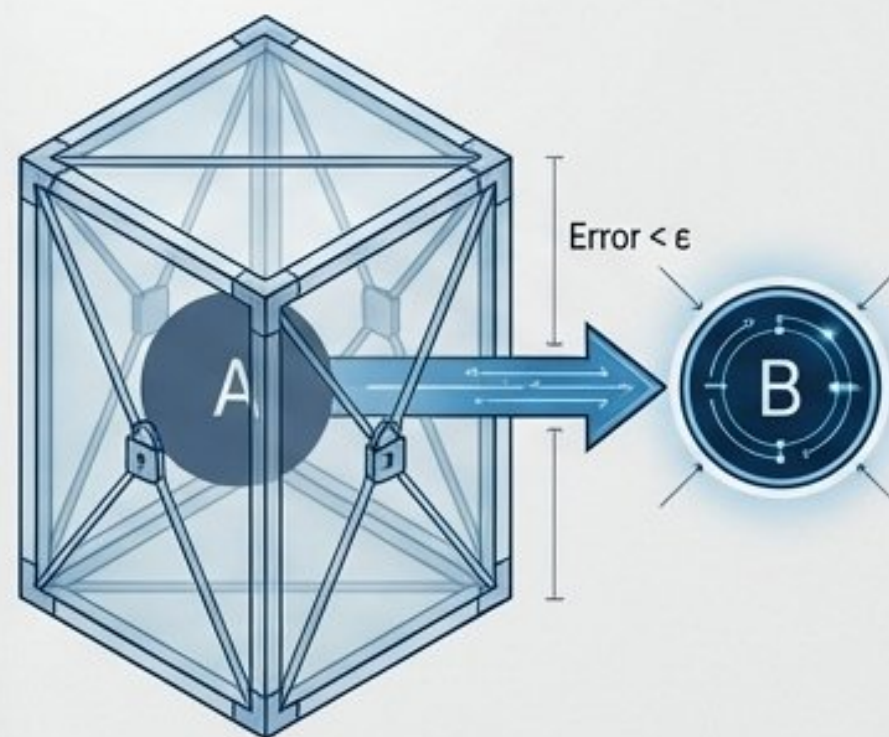
Test 1: Exclusivity

Remove A. Does B disappear?
(Anomaly reduction ratio > 0.3).



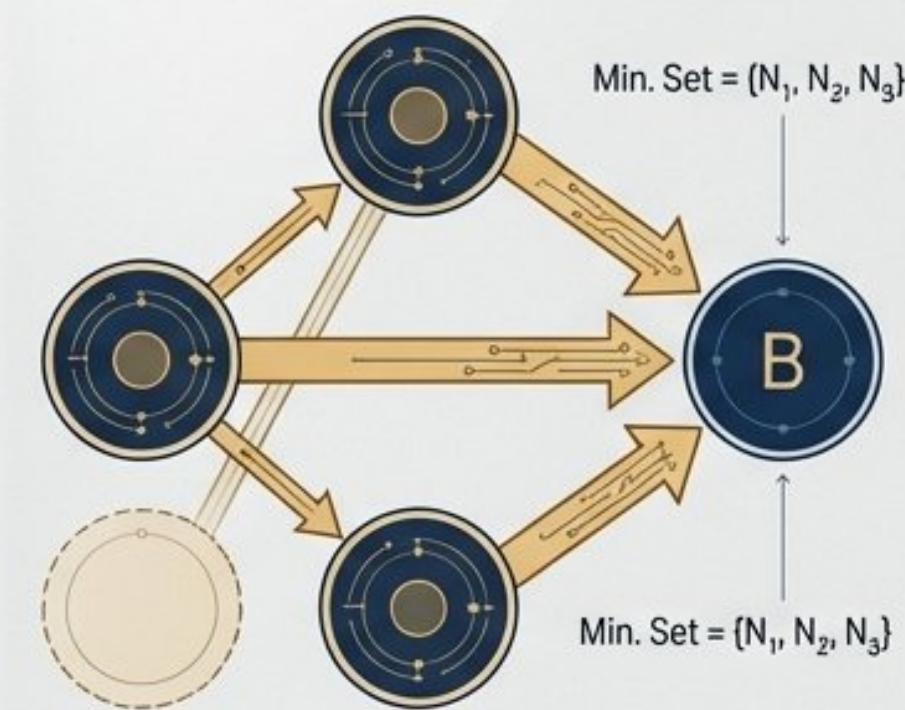
Test 2: Sufficiency

Isolate A. Does it accurately predict B?
(Prediction accuracy < 0.5 error).



Test 3: Minimality

Iteratively remove nodes from a cluster to find the minimal sufficient set.

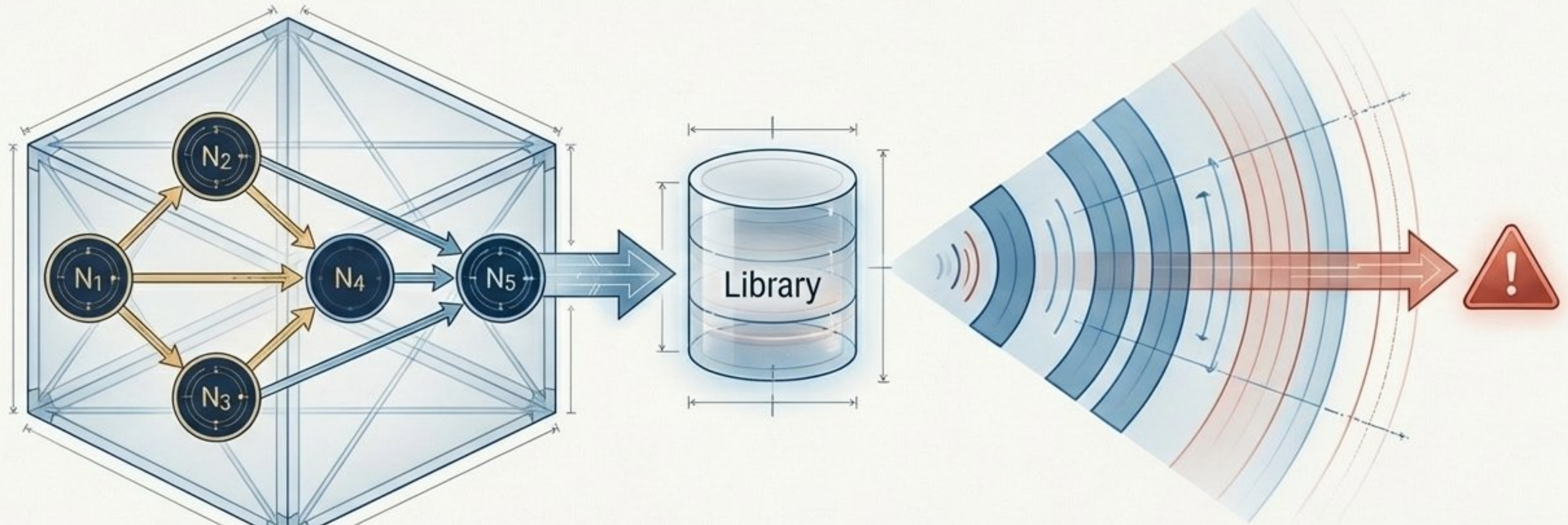


Outcome: Bounding Type I errors (false causal claims) at $\alpha = 0.05$ with 1,000 bootstrapped iterations.



Synthesizing causal graphs and predicting future cascades.

Module 6: Causal Insight Graph Synthesizer (CIGS)

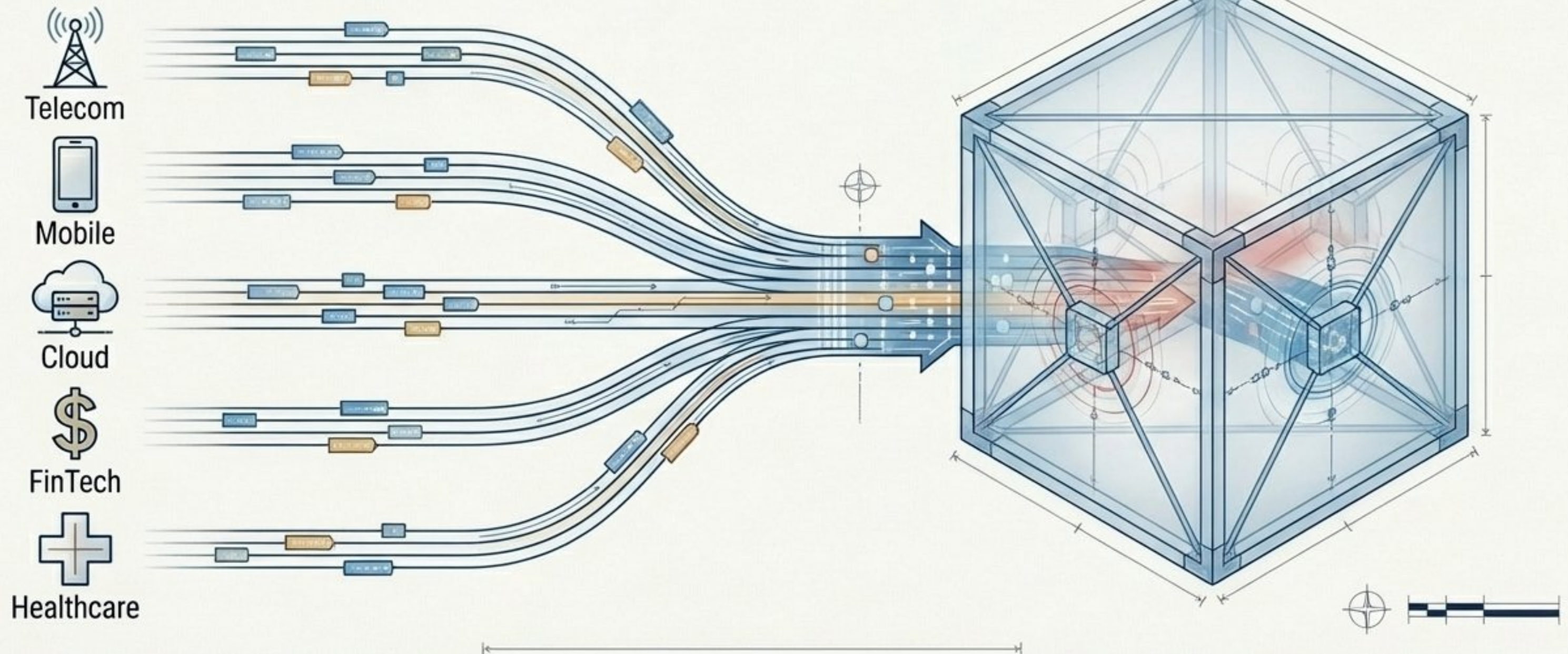


Assembles validated correlations into a causal chain, outputting the root cause, propagation nodes, terminal effects, and aggregate confidence score.

Module 7: Predictive Precursor Engine (PPE)
Matches new DST patterns against historical causal graphs to issue early warnings before systemic failure manifests.

A domain-agnostic engine requiring zero manual configuration.

Because AMACE maps all incoming data to the universal 3-stratum model and translates all anomalies into the algebraic DST format, the core methodology functions optimally regardless of the underlying domain terminology or data type.





Resolving cascading failures

Telecom Network

Correlated S0 firmware update with S1 latency and S2 churn risk.



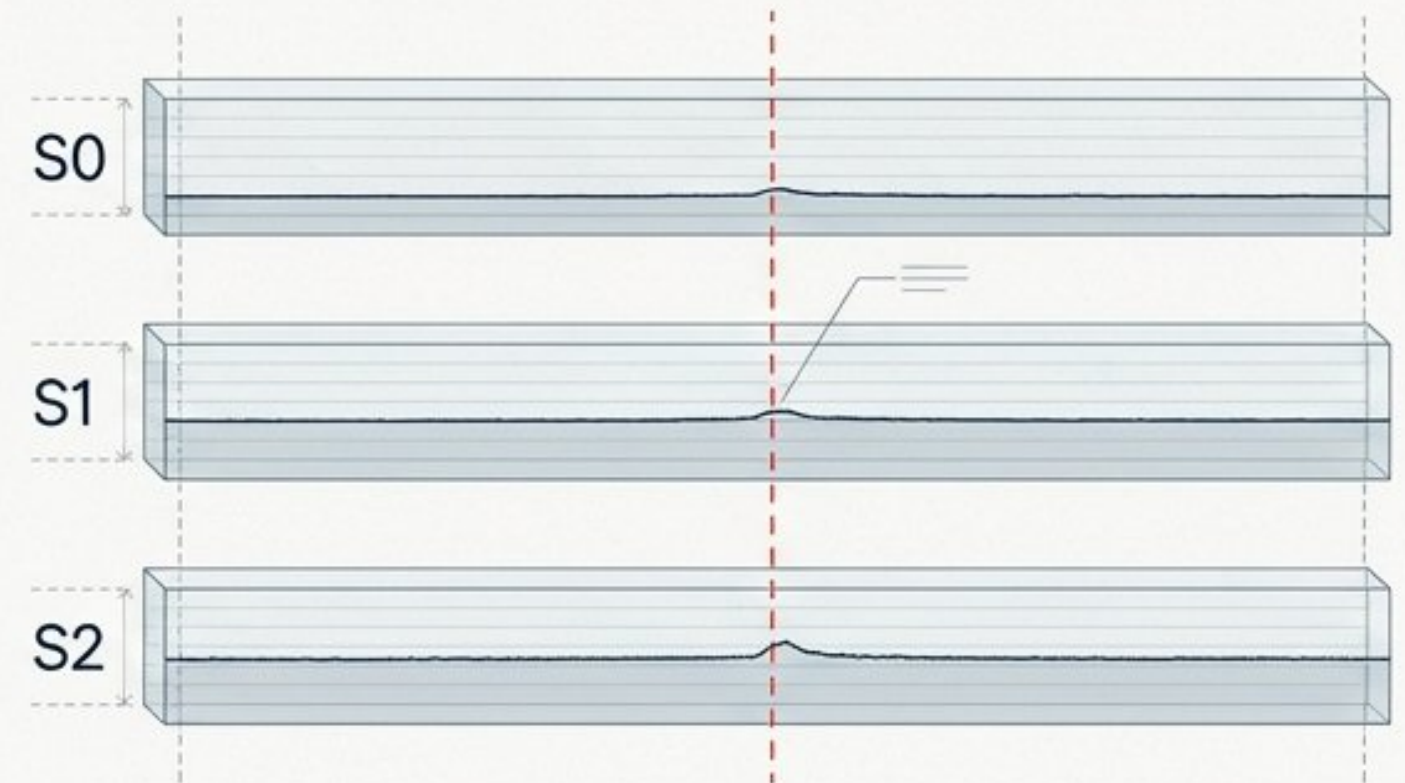
Time-to-root-cause reduced from 14 hours to 4.2 minutes



Surfacing silent revenue loss

Mobile App Platform

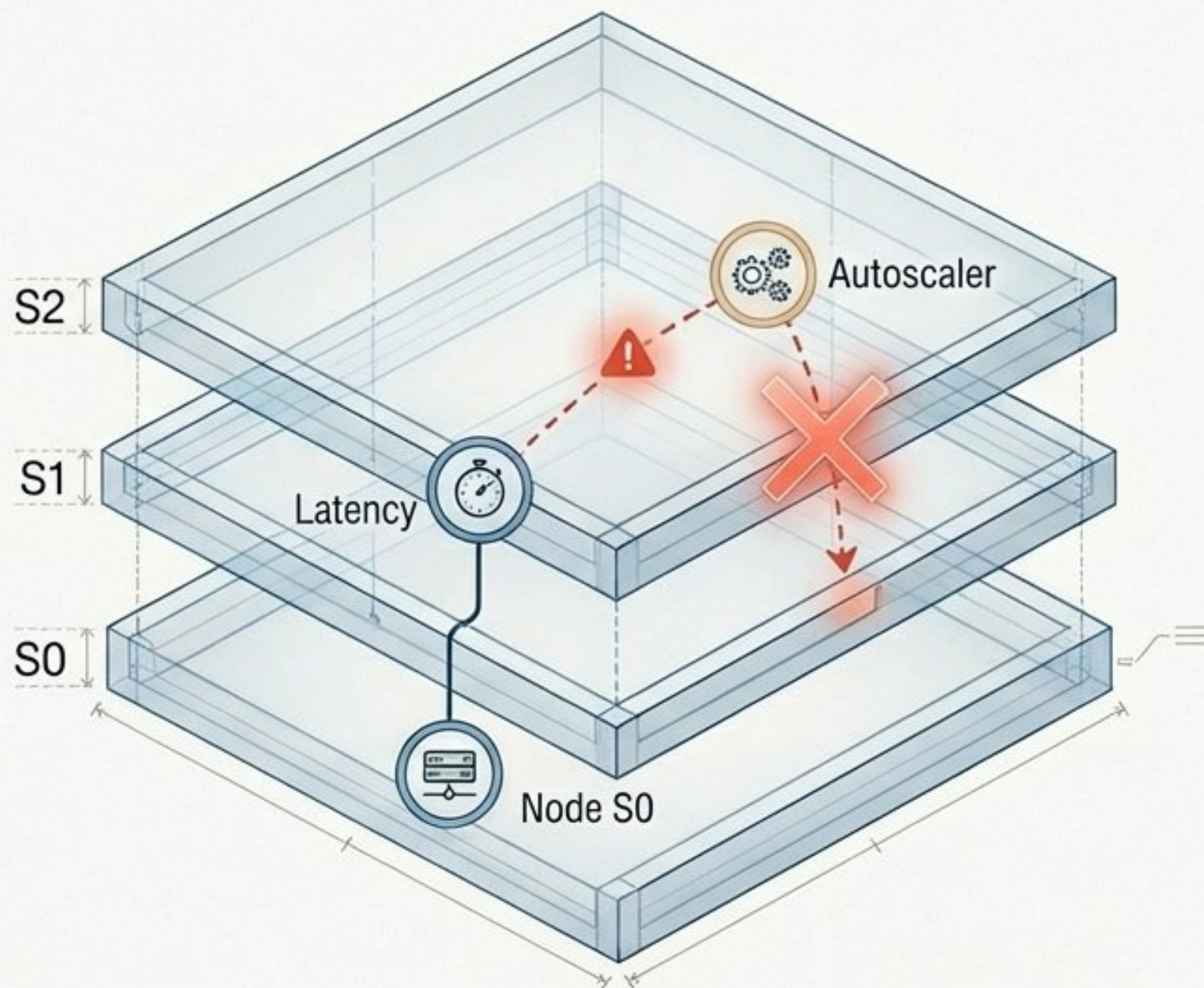
Caught a subtle API latency increase that breached no individual system thresholds, but sequentially caused render delays and CTR drops.



Prevented a hidden \$2.4M/day revenue bleed

Anti-correlation detection isolates latent infrastructure bugs.

Cloud Infrastructure Cascading Failure



The Missing Link:

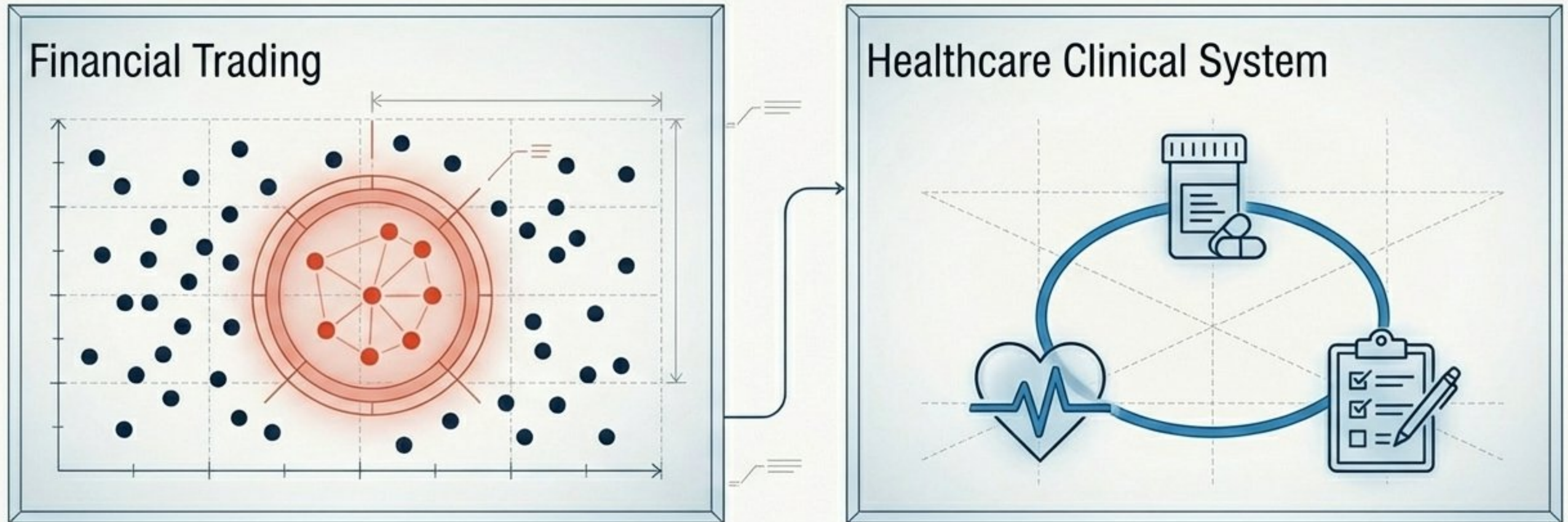
AMACE detected that the autoscaler **SHOULD** have triggered when latency breached, but did not.

Result:

Uncovered the root cause memory leak **AND** exposed a previously unknown, silent autoscaler misconfiguration in 8 minutes.



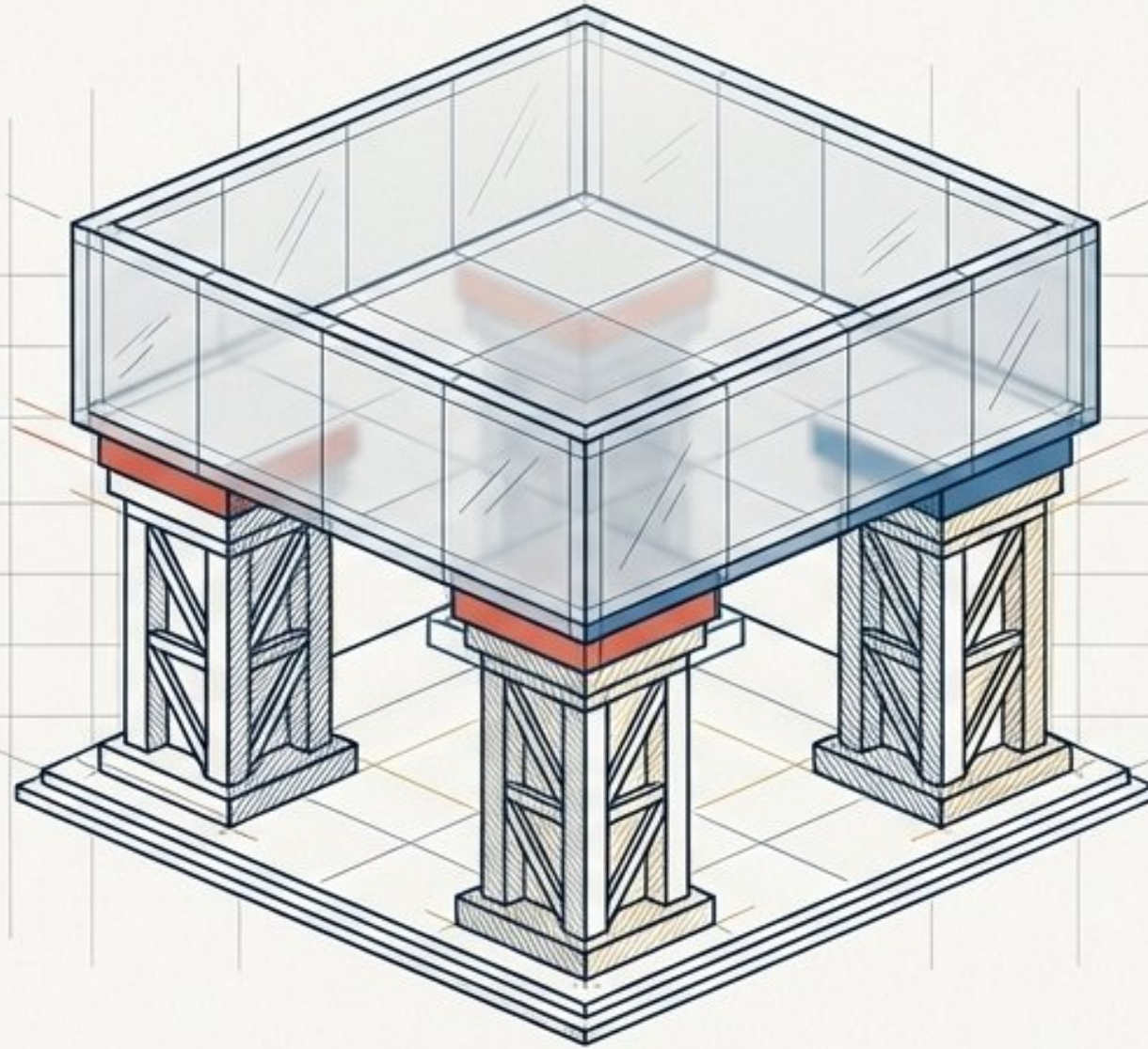
Distilling systemic signals from chaotic domain noise.



Separated genuine systemic risk exposures from coincidental market volatility. Filtered 47 independent risk alerts down to 7 actual correlated exposures, preventing \$4.7M in unnecessary liquidation.

Linked pharmacy overrides, elevated vitals, and degraded readmission models to detect a cardiac medication dosing error in 48 hours, eliminating an 11-day manual detection gap.

A specification engineered to survive systemic patent challenges



101: Alice

Concrete data structures (DST) and an exponential reduction in computational complexity proving improvements to computer functionality.

103: Obviousness

Unexpected results, including anti-correlation detection and 99.97% mathematical pruning efficiency.

112: Enablement

Complete mathematical definitions, algorithmic complexity bounds, and exact parameter specifications for implementation.

Defeating Abstract Idea and Obviousness arguments.

Attack Vector

This is just correlating information.

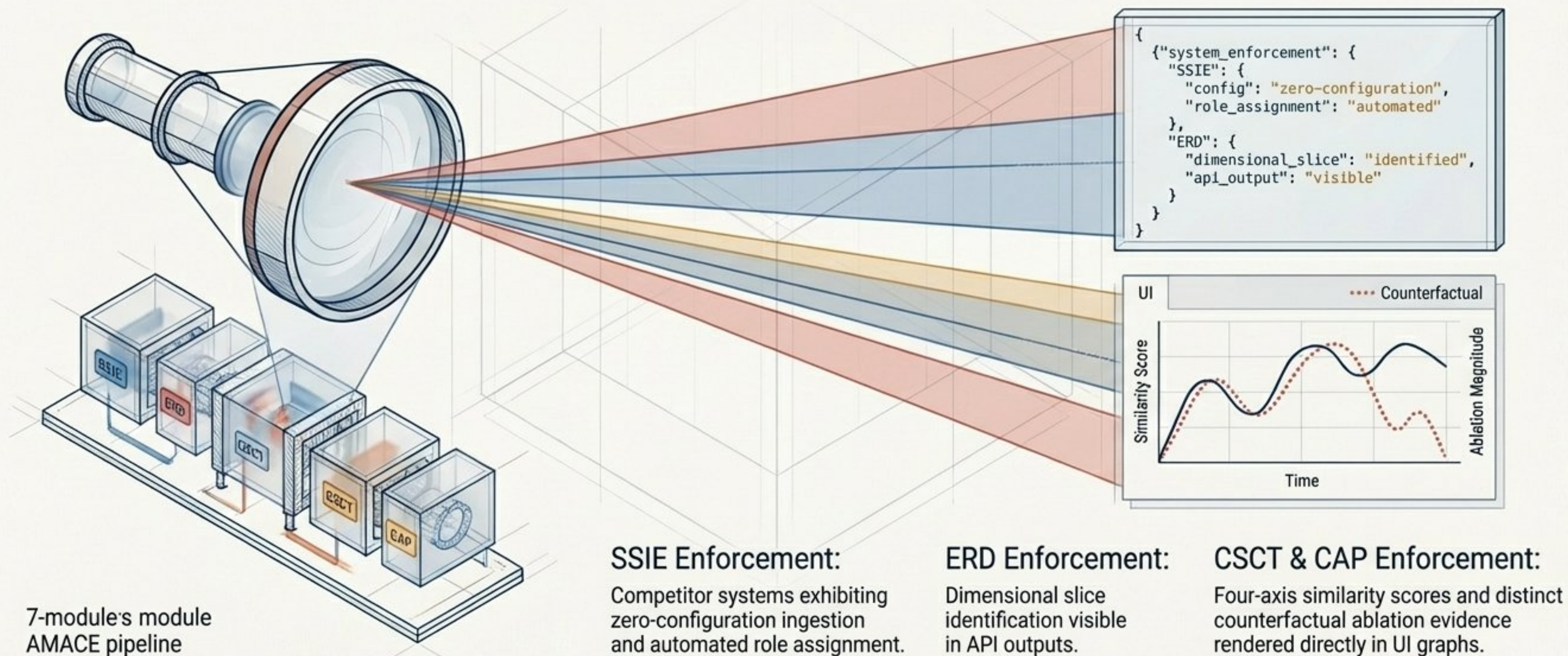
Predictable results from known elements.

AMACE Defense

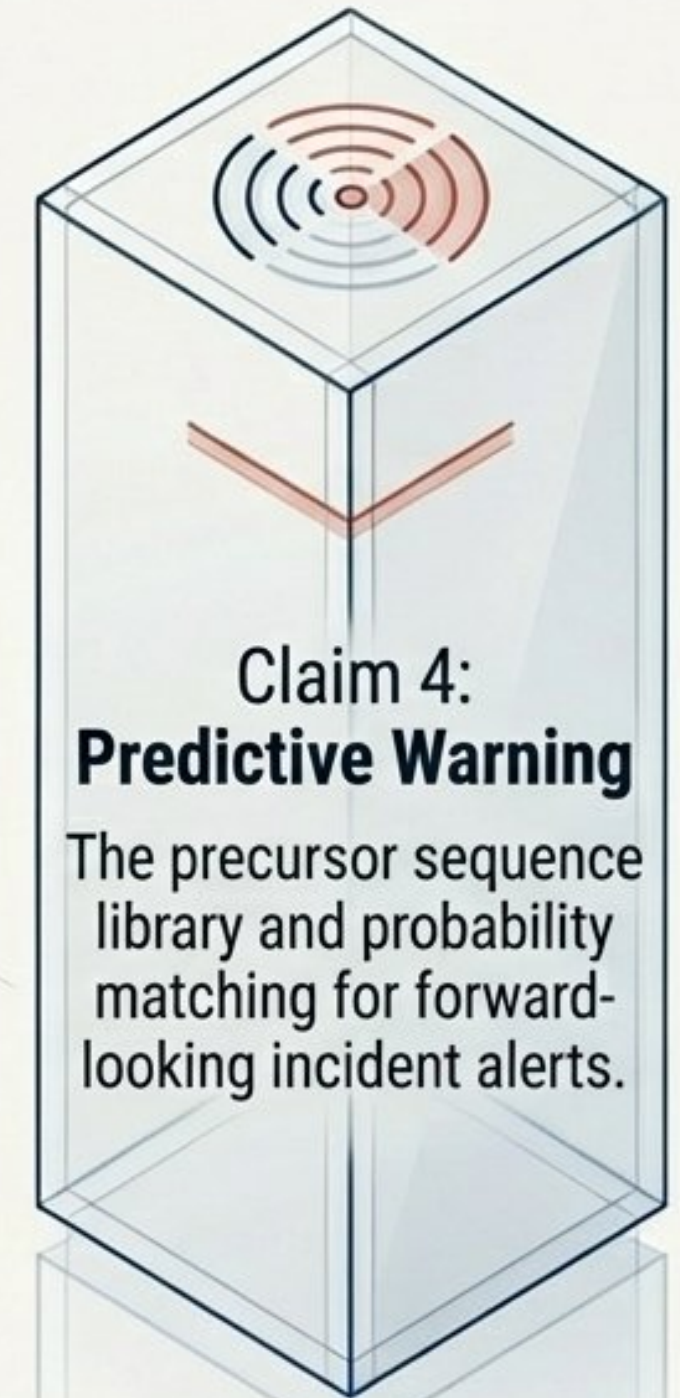
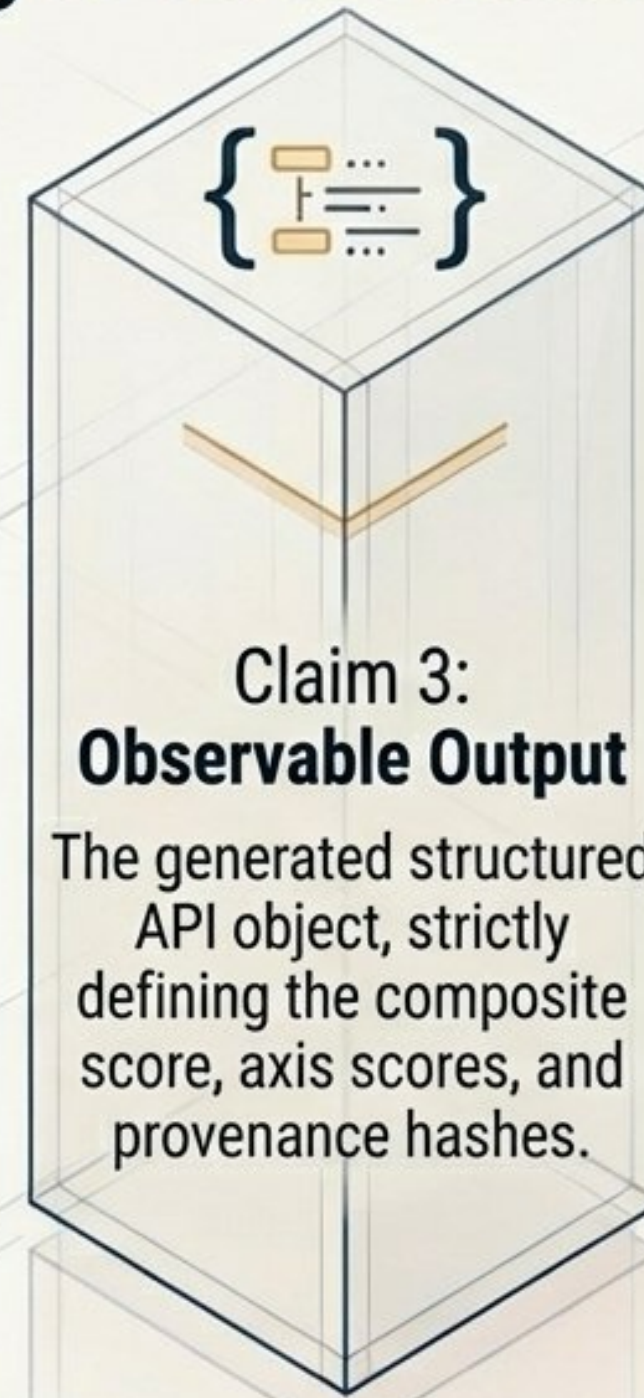
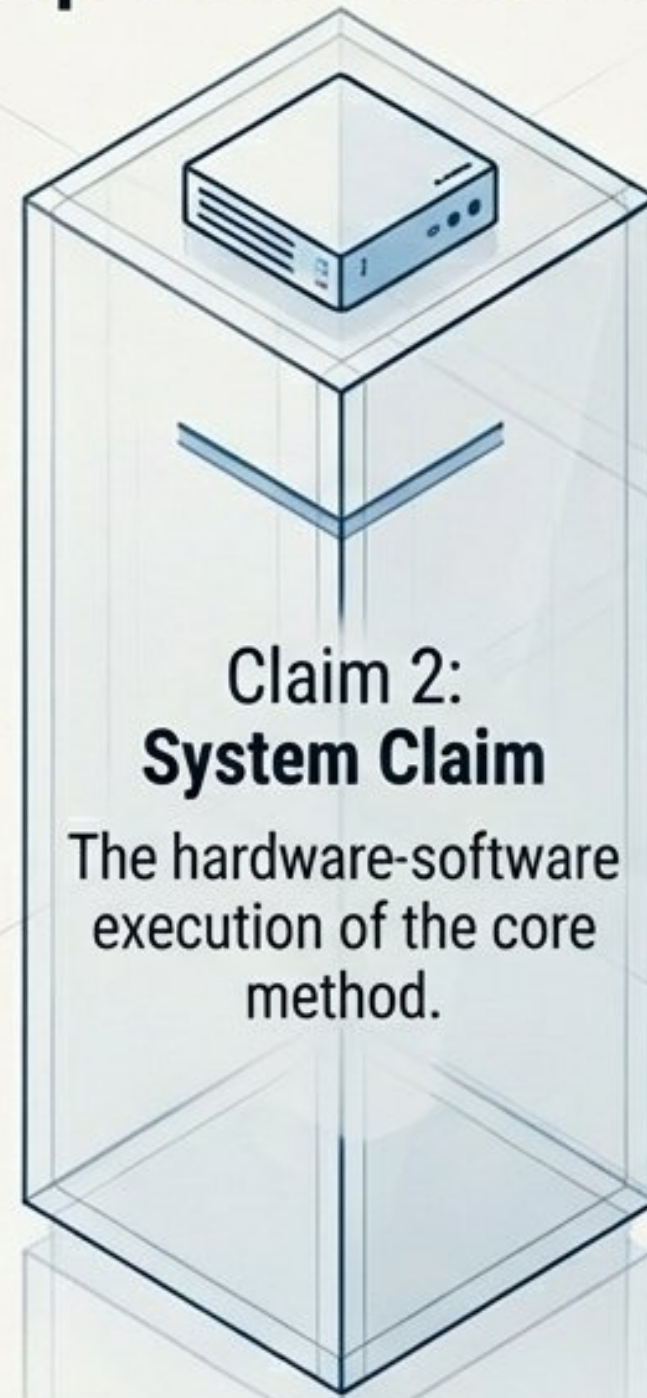
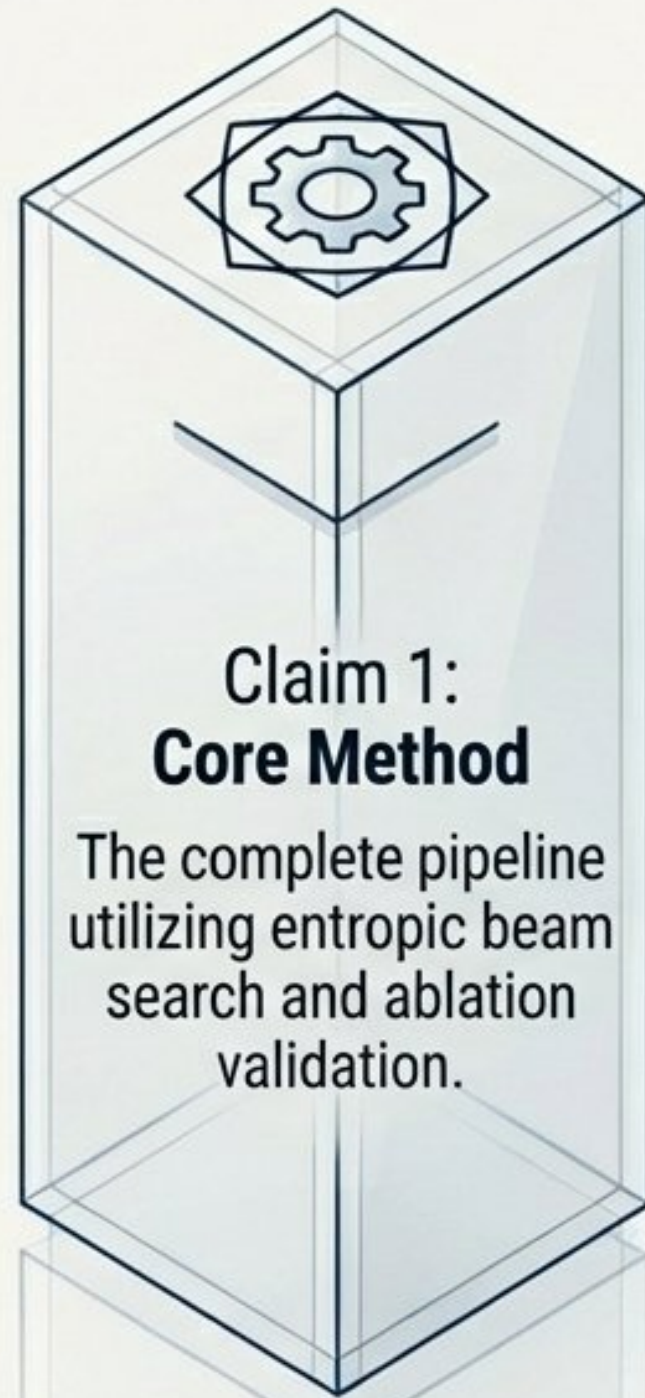
AMACE utilizes non-generic computational operations, specifically bootstrapped ablation with 1,000 iterations—a fundamentally machine process impossible for humans.

15+ years of prior art explicitly teaches away by isolating strata. AMACE produces unpredictable outcomes, such as surfacing sub-threshold anomalies solely through multi-layer alignment.

Internal algorithmic rigor produces externally observable artifacts.



Four independent claims form an impenetrable legal scaffolding.



The cross-stratum blind spot permanently resolved.

AMACE transforms chaotic, multi-layer heterogeneous data into counterfactually proven causal intelligence.

- Sub-quadratic complexity.
- Zero-configuration ingestion.
- $O(N \log N \times k \log D)$ efficiency.
- Provable causality.

